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REVIEW ARTICLE

From GIS to neogeography: ontological implications and theories of truth

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Neogeography has emerged as a descriptive and analytical tool for large numbers of people outside of academia, a process catalyzed by digital mapping technologies and the social networking practices of Web 2.0. This article examines the ontological and epistemological implications of this transition. It argues that neogeography has democratized GIS practices, facilitated the adoption of relational views of space and place, and broadened the ontological scope of GIS. Neogeography also poses epistemological challenges to the dominant theory of truth, in particular advancing a shift away from the correspondence model of truth toward consensus and performative interpretations. While recognizing the multiple unintended consequences of neogeographical practices, neogeography is held to be as a useful means for charting geographic space in light of intense postmodern time–space compression.

Keywords: neogeography; ontology; theories of truth; relational space; crowdsourcing; volunteered geographic information Web 2.0

Introduction

The dazzling, sometime even dizzying, development of new technologies for handling geospatial information has caught everybody by surprise over the past decade, particularly in the last 5 years. Neologisms burst into our lexicon at an overwhelming rate: Web 2.0, user-generated content, participatory GIS, 'do-it-yourself' cartography, MapQuest, Bing maps, location-based services, cloud computing, wikification, Google Earth, blogging, Flickr, digiplaces, YouTube, Geocaching, Second Life, augmented reality, Twitter, and Foursquare. All of these phenomena point to a profound shift in the nature and role of geographic information, a transition characterized by a 'bottom up' reconfiguration in how data are collected, transmitted, analyzed, visualized, and utilized that differs considerably from traditional 'top-down' models in which experts and government agencies dictate the criteria of data collection, analysis, applications, and standards of truth.

These diverse geographical practices have been loosely called neogeography (Turner 2006), a term coined by Di-Ann Eisnor of Platial (www.platial.com) (Walsh 2008). As Haden (2008) noted, the term has been used multiple times at major technological or epistemological junctures in geography throughout the twentieth century. We adopt both a narrow and a broad definition for neogeography (Vander Wal 2005). A narrow view refers to the process whereby varied groups of people use an eclectic set of online geospatial tools to describe and document aspects of their lives, society, or environment in terms that are meaningful to them (Hudson-Smith *et al*. 2009). Defined this way, neogeographical knowledge emanates from personal interactions with places, and what is held to be true is often highly contextual and specific to the community of users who generate such data. In a sense, then, neogeographical knowledge is inseparable from the rhythms and contours of everyday life. Rana and Joliveau (2009, p. 75) argue that to fully appreciate the significance of this phenomenon, neogeography must be regarded 'as an extension of mainstream geography for everyone made by everyone.' A broader interpretation of neogeography includes the cultural mapping of all realms of everyday life, catalyzed by the digital mapping technologies and social networking practices of Web 2.0 (Barnes and Sheppard 2010). As Goodchild (2009a) points out, neogeography often implies that academic geography is redundant or unnecessary. For this reason, he instead advocates terms such as volunteered geographic information (VGI).

In many ways, as demonstrated by the papers presented during the 2010 neogeographers' annual meeting, the Where 2.0 Symposium (http://en.oreilly.com/where2010), neogeography, code spaces, and the automatic production of space have become deeply woven into the fabric of daily life for countless millions of people to such an extent that simple dichotomies like 'online' and

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'off-line' fail to do justice to the ways in which the real and the virtual worlds are interpenetrated (Thrift and French 2002, Dodge and Kitchin 2005, Gordon 2008, Messinger *et al*. 2009). Although this phenomenon has been the subject of much recent scrutiny (Goodchild 2007a, b, 2009b, Elwood 2008, Coleman *et al*. 2009, Coote and Rackham 2009, Batty *et al*. 2010, Haklay 2010), the ontological and epistemological implications have remained largely unexplicated. Although one philosophical assessment has been offered about Web 2.0 and the semantic web in general (Floridi 2009), as yet there exists little systematic theoretical scrutiny of neogeography, which is supported by Web 2.0 technologies and the development of the semantic web. The aim of this article is to clarify the complex, multifaceted implications of neogeography for the production of geographical knowledge in general, and theories of knowledge and truth in particular, at both the ontological and epistemological levels. In an age in which theoretical self-consciousness and sophistication have reached unparalleled heights among critical social theorists, GIScientists can ill-afford to view the creation of 'truth' unproblematically when it has increasingly come to have a radically different meaning from the conventional academic one.

The article consists of four parts. It begins with a recapitulation of critical human geography's objections to conventional GIS, that is, its obsession with mechanical rationality, its denial of an observer and reliance on the correspondence theory of truth, its assumption that representation is only a technical, not social process, and the view of absolute space embedded within such approaches. Second, it turns to the rise of neogeography and critical GIS and its implications for data/information gathering and ultimately for knowledge production. We highlight how in practice neogeographers use geospatial technologies in multiple ways that are consistent epistemologically with the views of critical GIS scholars. Third, it discusses the pluralist ontologies that neogeography espouses, that is, the multitudinous realities perceived and inhabited by different social groups, with particular emphasis on representing relative and relational space. Fourth, we examine new approaches to truth and how these different approaches can be applied to neogeographies, namely, ways of understanding space that are free from the unrealistic constraints of the correspondence theory of truth, drawing on the works of three philosophers, Paul Feyerabend, Jürgen Habermas, and Richard Rorty. We discuss how neogeographical practices are consistent with alternative theories of truth such as the consensus or performative theories of truth. The last section offers a conclusion that situates neogeographic practices in the context of fractalized, postmodern life.

Conventional versus critical GIS

To appreciate neogeography's implications as a new mode for geographical knowledge production, a quick review is offered to discuss the critiques of conventional GIS and how neogeography reflects the broader rise of critical GIS. Following a long tradition of critical cartography (Harley 1989, Wood 1992, Crampton and Krygier 2005), it has become common for geographers to view maps as much more than simply objects, but as configurations of power and knowledge that are indelibly linked to the social relations that produce them. Such an epistemological shift challenges conventional views of representation as a purely technical act and recasts it as one intimately tied to social interests, that is, with social origins and social consequences. In this light, the traditional boundary between epistemology (the theory of knowledge, i.e., how do we know?) and ontology (the theory of reality, i.e., what exists?) dissolves. Maps as a form of spatial knowledge do much more than simply reflect the world, but in fact discursively constitute it, or as Wood (1992) notes, they help to bring into existence the world that they portray. Seen from this perspective, maps – and geospatial technologies in general – lose much of their conventional aura of objectivity and scientific detachment and are recast as social products with social origins and social consequences.

Conventional GIS is essentially spatially deterministic in nature in that it requires representations of landscapes and processes to be tied to the geometrical primitives of points, lines, polygons, and pixels. The mathematical topology that underpins GIS brings its own data representations in the form of raster, vector, and object forms. The deployment of these geometric forms enables conventional GIS to classify readily observable empirical phenomena such as natural resources, infrastructure, demographic patterns, and environmental events rather than use the less well-defined descriptive terms and categories deployed in critical geography and social theory such as class, ethnicity, gender, and other markers of social identity. Because technologies shape how we view the world, conventional GIS inevitably rested upon, and amplified, an essentially positivist philosophical perspective at the expense of others. Epistemologically, conventional cartography and GIS tend to privilege the quantitative and the observable over the qualitative and the nonobservable. Reality is therefore assumed to consist of only what is seen or measurable, and it is things themselves, not the relations among them, which hold ontological priority (a notion that pervades both positivism and empiricism). Such a perspective upholds a deductive, nomothetic rationality that marginalizes other logics such as social constructivism or hermeneutics. Conventional GIS unsurprisingly tends to value precision, simplification, certainty, and deterministic causality over complexity, ambiguity, multiplicity, and contingency, the very things that most heavily concern critical social theorists.

Epistemologically, conventional GIS relies exclusively on the correspondence theory of truth, particularly the version formalized by the Vienna Circle in the early twentieth

century, in which the truth value of theoretical statements is judged by their isomorphism with observed facts, that is, the correspondence between hypotheses and data. Truth, in this view, unproblematically exists 'out there,' independently of whether or not anyone believes in it or not, which is to say that truth is held to be not a social construction but an objective, given fact. In the correspondence model, truth is 'real' whether or not anyone believes in it. Such a perspective subscribes to a strict dichotomy between theoretical and empirical languages (i.e., the data are assumed not to be theory-laden: facts are given, not made), renders the role of the observer invisible or irrelevant, and has long been dominant not only in Anglophone analytic philosophy but in much of the social sciences as well. This notion confers the power to know and define the world – both social and natural – on an invisible, rational, implicitly male, decontextualized, all-knowing observer – the Cartesian cogito. However, in the face of intense criticisms, many social scientists, and practitioners of critical GIS, have abandoned the strict positivist view of the world, although it continues to dominate the physical sciences.

In addition to its views of knowledge and the knower, conventional GIS also relies upon a particular notion of space. Specifically, like the isotropic plains of location theory and regional science, traditional practitioners of GIS invariably invoke an absolute Cartesian view of spatiality as given, not produced, a homogenous surface devoid of social origins. Absolute space – infinite, absolute, and homogenous – has a long lineage in Western history that can be traced back to classical Greece, including geometry. Euclidean geometry, grounded in the assumption of one, uniform, continuous space, dominated the mathematics of spatial representation for the next two millennia. In Renaissance Europe, Cartesian views of space were instrumental in the development of cartography, land surveying, civil engineering, and architecture (Cosgrove 2008). The Cartesian/Euclidean notion of space was replicated throughout the Renaissance and the Enlightenment and formed the basis for Newtonian physics. Such a perspective was instrumental to the emerging capitalist logic of globalized commodity production and consumption. For European navigators, for example, smoothing space by reducing it to distance rendered the oceans navigable, and arranged the multitude of the planet's places into a single, unified, coherent, and panopticonic understanding of the world. In the United States, the township and range system played a similar role. In the process, space was robbed of substantive social context to become an ordered and uniform system of abstract linear coordinates (e.g., the grid of latitude and longitude).

The social constructivist critisms of conventional cartography found their way into GIS in the 1990s, with dramatic results (Pickles 1995, Sheppard 1995, 2005a, b, Schuurman 2000, 2006). Because trenchant critiques are found elsewhere, there is no need to recapitulate this literature here. Suffice it to say that critics held that conventional GIS lacked a sufficient engagement with issues of social context, relations, and power, amounting to little more than a Trojan horse for smuggling empiricism or positivism back into geographic practice. Others were concerned about the military and corporate origins and applications of much of GIS, potential invasions of privacy, and the panopticonic implications of centralized geospatial databases (Armstrong and Ruggles 2005, Dobson and Fisher 2007). The community of conventional GIS practitioners initially amplified the schism with social theorists through its impoverished epistemological self-understanding, which was typically framed in a purely technocratic language of engineering, efficiency, control, and manipulation, as if GIS were devoid of social roots and consequences.

However, over time, in response to these criticisms, the GIS community has changed steadily. Indeed, there is no reason why GIS cannot 'be employed within non-positivist epistemologies' (Pavoloskaya 2002, p. 287). The evolution of GIS since the 1990s reveals a history that currently includes a fantastic zoo of new species, including, to take but a few examples, critical GIS (Sheppard 2005b), participatory GIS (Elwood 2006), volunteered geography (Goodchild 2007b), affective/emotional GIS (Aitken and Craine 2009), qualitative GIS (Cope and Elwood 2009), feminist GIS (Kwan 2002a, McLafferty 2002, 2005, Story 2002), queer GIS (Brown and Knopp 2008), digiplaces (Zook and Graham 2007a), ethnographic and indigenous knowledge (Robbins 2003), and humanistic GIScience (Sui 2004, Bodenhamer *et al*. 2010). These developments reflect a highly productive engagement between the GIS community and social theorists and humanities scholars during the last decade. Interactions and exchanges between GIScientists and social theorists stimulated the emergence of so-called critical GIS that attempts at 'pixelizing the social and socializing the pixel' (Geoghegan *et al*. 1998). Although there are certainly significant differences among these diverse approaches, there are also numerous similarities that wind their way through this body of work. Thus, contemporary forms of GIS, unlike those dominant in the 1990s, tend to be much more context-sensitive and issue-driven than their predecessors. In addition to paying more attention to issues of power, commodification, and surveillance, GIS practices nowadays are also more likely to tolerate higher levels of fuzziness, uncertainty, and lack of precision and accuracy. Unlike conventional GIS, local communities are intimately involved in the production of neogeographic GIS information, and indigenous forms of knowledge are accepted as valid and important sources of information. With more emphasis on cooperation and collaboration, contemporary GIS practices are considerably more sensitive to issues of access, control, and ownership of spatial information (King 2002, Dunn 2007). The feminist critique of GIS (Schuurman and Pratt 2002),

for example, reveals both the masculinist bias of conventional GIS and points toward an alternative that is more empathetic and caring in nature by incorporating qualitative measures of emotional states such as fear (Kwan 2002b). Neogeography may be seen as yet one more step in this evolution, initiated by a crowd of citizens who are empowered by Web 2.0 and social networking technologies (*Economist* 2010).

Although these practices have substantively altered the ways in which GIS is practiced, we concur with Leszczynski (2009a) that there still remains a deep 'disconnect' between critical human geography's critiques and many GIS practices. Calls have been made for a critical engagement between social theorists and the community of GIS users on the basis of a common material ontological ground, that is, on the basis of the technology's applications and not on the disparate, academic linguistic terms that divide practitioners of conventional GIS and social theorists (Leszczynski 2009b, c). We argue that this disconnection is gradually being overcome by neogeography, thus generating more pluralist ontologies in the age of Web 2.0.

Neogeography and its consequences

During the past several years, neogeography has enjoyed explosive growth (Turner and Forrest 2008), to the point that Turner (2008) claimed that neogeography may even have assassinated GIS. Neogeography both resembles and departs from close kin such as participatory GIS. Neogeography enables extensive, active bilateral contacts between the producers and analysts of geographic data, resulting in a constantly mutating body of work, usually without a clearly defined political agenda. In contrast, participatory GIS tends to confine local and popular input to the generation of data often tied to controversial issues with explicit political implications, such as local zoning disputes, siting of toxic facilities, and so on (Dunn 2007).

At the core of the shift into neogeography is Web 2.0, the set of software revolutionizing applications on the web. Key to the architecture of this technology are Asynchronous Javascript And XML (AJAX) and Application Programming Interfaces (API), which allow instantaneous user interactions. The functionality offered by Web 2.0 has precipitated significant changes from traditional approaches. Whereas conventional GIS is expert-centric, neogeography is obviously user-centric. Most neogeographic practitioners, who harness the technology for their own purposes, often have limited technical skills, and typically care little about academic debates regarding GIS. In this sense, neogeography has helped to foster an unprecedented democratization of geographic knowledge, often with roots far removed from academic experts (Couclelis 2003). Thus Goodchild (2007a, p. 27) maintains that whereas 'the early Web was primarily one-directional, allowing a large number of users to view the contents of a comparatively small number of sites, the new Web 2.0 is a bi-directional collaboration in which users are able to interact with and provide information to central sites, and to see that information collated and made available to others.' The interactive websites supported by Web 2.0, in contrast, allow users to upload locations into online content and apply their data in diverse ways, including, for example, simple displays of locations (e.g., favored bird-watching sites) or lists of attributes of a place near a user equipped with a GPS. This approach lies at the heart of mapping websites such as Google Maps, Yahoo!Maps, OpenStreetMap, and Bing Live Maps. Unlike traditional GIS approaches, users can create, share, and use information through 'crowdsourcing' (Howe 2008), which allow large, widely distributed groups of people to work together toward a common goal. For example, Jackson (2010) reported that OpenStreetMap has attracted 200,000 volunteers annually since its inception in 2004. Similarly, Google Maps was used by 71.5 million people in 2007 and Google Earth by 22.7 million (Haklay *et al*. 2008). Indeed, rather than relying on state or corporate-produced data through a top-down process, neogeography generates volunteered data and content produced by citizens via a bottom-up process, which has relocated the center of knowledge production from a handful of self-appointed experts to large numbers of people with limited formal geographic training. Sui (2008) labels these changes the 'wikification of GIS,' after Wikipedia, the famously popular, user-generated, online encyclopedia. Wikification represents a significant step forward in the democratization of geographic information, shifting control over the production and use of GIS data from a handful of experts to large groups of users. As a consequence, GIS has increasingly escaped the lonely confines of academia or planning professionals and has been embraced by an enthusiastic and rapidly growing public of amateurs and hobbyists. GIS, it seems, is too important to leave to conventional GIS practitioners.

As a result of its popularity, neogeography's impacts have multiplied, penetrating the natural sciences, social sciences, health care research, and the humanities. Examples include applications in physical geography (http://www.climatecollaboratorium.org), human geography (floatingsheep.org, www.popvssoda.com; Elwood 2010), geographical education (Moulder 2008, Papadimitriou 2010), tourism (Girardin *et al*. 2008), public health (Boulos *et al*. 2008), and emergency management (Liu and Palen 2010). Neogeography was instrumental in crafting responses to the recent Haitian earthquake and British Petroleum's disastrous oil spill in the Gulf of Mexico (Zook *et al*. 2010). Neogeography has also found uses in political geography (Shin 2009; www.fairvote2020.org), popular place descriptions (Dykes *et al*. 2008), and spatial decision making (Rinner *et al*. 2008), all of which fall under the broader umbrella of citizen science (Cohn 2008).

Neogeography, broadly speaking, can be regarded as part of the emerging citizen science, with VGI as its primary input source enabled by the geospatial web (or GeoWeb), that is, the location-based, geocoded data rather than the aspatial information currently hegemonic on the Internet (Scharl 2007).

Although the diverse motivations of people who make contributions to neogeography sites still need further clarification (Cook 2008), neogeography bears distinct similarities to parallel lines of thought such as complexity theory, agent-based modeling, and complex adaptive systems, all of which have also found their way into the academic discipline of geography (O'Sullivan 2004, Manson and O'Sullivan 2006, Batty 2007). Despite these manifest differences, there are also important commonalities. In both neogeography and other approaches, the analytical focus is 'bottom-up,' that is, on the relations among large numbers of actors (rather than the individual actors themselves), who, often inadvertently, create order out of chaos (a view that also resembles Giddens's structuration theory) and, sometimes, chaos out of order. Through the properties of emergence or self-organization, complexity theory implies that local spatial configurations of interactions affect outcomes at broader systemic scales, or what the literature on the social production of scale calls scale jumping (Smith 2005). Neogeographies are, in a sense, also complex adaptive systems in which conscious agents construct and reproduce systems of meaning, both intended and unintended. Neogeographic practices generally do not follow a trajectory of gradual and linear growth, but indeed often burst into existence suddenly when a critical mass of followers adopt them quickly. Such a development pattern is consistent with the discoveries of the emerging science of networks (Barabasi 2010). Neogeographic systems generate qualitatively more value as their size and complexity grow. Analogously, according to Metcalfe's law (Hendler and Golbeck 2008), the value of a telecommunication network to each user is proportional to the square of its total number of users. For the geospatial community, the growth from one million GIS users to 400 million neogeographers worldwide is a game-changing shift.

Although neogeography has frequently been celebrated in the popular press in rosy, utopian terms, it is well worth noting that it also suffers from problematic concerns of its own (Kingsbury and Jones 2009). Five issues are noted here. First, simplistic worship of 'crowd wisdom' can easily slip into uncritical reliance upon groups of uninformed amateurs, with potentially disastrous results, or 'crowd stupidity' (Keen 2008, Sunstein 2009). User-generated content tends to rely on polarizing samples of data – only those who either love or hate the subject matter beyond a threshold level tend to post their opinions online. These extremes can drown more subtle and accomodationist interpretations and generate conceptual and political discord (Sunstein 2009). For example, neogeography may amplify simplistic and erroneous stereotypes among selected groups of users, and downgrade the knowledge of experts (Keen 2008). Parks (2009) noted how Google Earth representations of the violence in Darfur often unproblematically reproduced unsophisticated Western notions about Africa. The popularity of Charles MacKay's (2010) classic *Extraordinary Popular Delusions and the Madness of Crowds* (originally published in 1841) further testifies to the public's interest in the mantra of crowd wisdom. Second, the glorification of open source software can exact a price in terms of individual creativity and issues of intellectual property rights, such as the music industry's notorious crackdown on file-sharing sites such as Napster. Third, despite its popular origins, neogeography is not free either from corporate attempts to commodify these technologies and the forms of knowledge they unleash (e.g., Google, Microsoft, Amazon.com). Fourth, there remains an important, largely positivist, epistemological objection: How reliable are the results of neogeographic approaches without random sampling of data, the lynchpin of scientific respectability (Bruns 2007, 2008, Sunstein 2009)? Self-selected groups of users inevitably will create pools of information with profound biases that reflect their social and spatial positionality. For example, the biases of VGI are particularly acute at the global level due to the digital divide (e.g., half of the world's population has not made even a single phone call, and only one-quarter has access to the Internet). Fifth, neogeography also serves as a potent reminder of the militarization of everyday life by geolocating precisely where everything and everybody is situated on a 24/7 basis (Kaplan, n.d.). In a world in which powerful corporate and state interests rely upon accurate geocoded data for often unsavory purposes of surveillance and control, VGI may enhance, not reduce, power differentials in the access to and production of geographic knowledge. Perhaps more than ever, we need a broader dialogue on the legal and ethical implications of pervasive penetration of geospatial technologies into our daily lives (Sui 2010).

Neogeography and alternative conceptualizations of space and place

As human geography shifted decisively into various forms of social theory and political economy, absolute space largely gave way to relative and relational conceptions of spatiality. The coincidence of this change with the rise of neogeography is far from coincidental; rather, relational space and neogeographic ways of collecting and representing data may be seen as simultaneously determinant. Just as conventional GIS carried with it a view of space as given and absolute, so too does neogeography offer the possibility of revisualizing space in relational and relative terms (although there are differences between relative and relational space, they are used here synonymously). In contrast to absolute space, which is fixed, asocial, and timeless,

relative and relational space reflects the wide varieties of ways in which distance is measured, perceived, and traversed, that is, space as socially made and remade over time. Rather than the long-dead geometries of positivism, relative space portrays geographies as fluid, mutable, and ever-changing.

The genesis of relative and relational space may arguably be traced to the famed seventeenth-century intellectual Gottfried Leibniz (1646–1716), Newton's rival and the founder of an alternative perspective on spatiality. The Leibnizian view of space is sharply at odds with the prevailing Euclidean/Cartesian emphasis on absolute space. In contrast to Newton, who held that space was absolute in the sense that it existed independently of how it was measured or what it contained, Leibniz held that time and space were comprehensible only with reference to specific frames of interpretation: distance, for example, could only be understood through the appeal of two or more objects situated in space. Relative space, however, remained subordinate to absolute space for many years, at least until the massive time–space compression of the Industrial Revolution reworked the meaning of spatiality itself, as wave upon wave of innovations reduced the friction of distance and led to a widespread reconfiguration of notions of proximity (Warf 2008). Relative and relational space became increasingly intertwined with the dynamics of modern and postmodern capitalism. For example, Kern's (1983) famous book *The Culture of Time and Space 1880–1918* addressed the changing meanings of spatiality in the midst of late nineteenth- and early twentieth-century technological and intellectual changes. The telegraph, telephone, bicycle, photography, automobile, airplane, impressionism, cubism, cinema, and theory of relativity, all blurred conventional meanings of near and far by bringing into contact peoples and societies that had long been isolated from one another. This transformation rendered traditional views of distance, and thus absolute space, less meaningful, and elevated the concept of space as a product of changing social circumstances.

Social theory proved to be decisive in initiating the shift from relative space to relational space, one with explicit social origins and consequences. Relational space, for example, found its way into Marx's summation of the process by which capitalism reached out and conquered new spaces, enveloping them within ever-broader spatial divisions of labor. Notably, David Harvey's (1990) famous spatialization of Marxism linked the construction and reconstruction of relational space to capital's endless search for ever greater profits and outlets for surplus value.

In the contemporary historical moment, globalized, postmodern capitalism has given birth to complex geographies in which proximity and transport costs play relatively marginal or even insignificant roles (Sheppard 2002). Airplanes, satellites, television, and telecommunications, including the Internet, for example, steadily led to forms of social interaction in which physical distance is largely rendered irrelevant, even if location and place are decidedly not. Consequently, we have witnessed a series of convoluted landscapes marked by wormholes and tunnels, giving rise to complicated, origami-like spatialities typified by rapidly changing geographies of centrality and peripherality. Geographic theorizations of relative and relational space include actor–network theory (Murdoch 2006), Massey's (1993 2005) power-geometries, and Deleuze and Guattari's (1987) 'schizophrenic' reconceptualization of rhizomes. All of these works point toward a topological conception of space quite at odds with that found in conventional GIS, that is, space as formed by contingent, rapidly changing, interconnected sets of networks in which relational connections among locales rather than their absolute positionality is the dominant characteristic.

Neogeography has injected relational space into the hitherto rather closed world of GIS. Egenhofer and Mark (1995) argued that formal models can be developed according to people's 'naïve geography,' commonsense knowledge about their surroundings that can serve as a means of incorporating daily life and lived experience into the design of GIS (Dourish and Bell 2007, Writer 2009). As cyberspace has become indispensable to ever larger numbers of people, including the 1.9 billion users of the Internet in June 2010 (www.internetworldstats.com/stats.htm), neogeography has allowed innumerable people, most of whom are outside of academia, to chart diverse paths through the virtual metaverse (Smart *et al*. 2007), the multiple spaces (Li *et al*. 2010), and 'digi-places' (Zook and Graham 2007b, Graham 2009) that are often populated by avatars (Papagiannidis *et al*. 2009). According to the report recently produced by the cross-industry public foresight project (www.metaverseroadmap.org), we are rapidly entering a new age of the metaverse – virtual worlds that serve as digital equivalents to the atom-based physical world. The browser for engaging this metaverse will be based upon a three-dimensional web that brings together the following four technologies:

- *Mirror worlds* – digital representations of non-virtual reality, such as Google Earth, Microsoft's Virtual Earth, NASA's World Winds, ESRI's ArcGlobe, and so on;
- *Virtual worlds* – digital representations of imagined worlds, such as Second Life and World of Warcraft;
- *Lifelogging* – the self-generated digital capture of information in the real or digital worlds, such as blogs, Twitter, YouTube, and social interactions via MySpace or FaceBook;
- *Augmented reality* – sensory overlays of digital information on the real and virtual worlds using head-up displays (HUDs) or other mobile/wearable devices such as cell phones or sensors through participatory sensing.

Viewed from the perspective of this emerging metaverse, traditional GIS has focused almost exclusively on georeferenced Cartesian coordinates defined by Euclidean geometry, a view that reifies absolute conceptualizations of space. In contrast, the diverse approaches grouped under the umbrella of neogeography include elements from all these four worlds that are linked together through heterogenous geo-tags, an approach in accordance with a relational view of space (Jensen 2010). Jones (2009) holds that relational thinking is a paradigmatic departure from the concerns of absolute space because it dissolves the boundaries between objects and space, that is, it sutures phenomena and context together into inseparably intertwined entities. Such a stance is firmly in keeping with the relational view of space, in which space does not exist as an entity in and of itself, over and above material objects and their spatial–temporal relations and tensions.

In addition to promoting an alternative view of space, neogeography also entails the incorporation of place with an emphasis on local information (Willis 2010), another conceptual shift that is quite different from traditional GIS practices. According to Tuan (1977), whereas space has often been associated by geographers with freedom, movement, distance, potential, and abstraction, place often implies confinement, stability, proximity, meaning, and the concrete. As Sack (1980) and Casey (1998) cogently argued, conceptualizations of space and place have a rich history and do not lend themselves to easy summary. Such a maneuver allows neogeographers to incorporate hitherto neglected data such as personal feelings at particular places and times (http://www.wefeelfine.org). Many neogeographic practices are in fact detailed descriptions of particular places or regions. Graham (2009) resurrected the once-popular palimpsest metaphor to describe what neogeographers have been contributing online: Web 2.0 technologies essentially serve as the new medium, similar to the papyrus- or parchment-based documents of an earlier era, for us to write and rewrite layers of information about the world. We need the concepts of both space and place to comprehend the ontology of neogeography.

Thus, rather than a single, uniform plain over which social relations unfold, the ontology of neogeography tends to be partial, overlapping, and specific to the interests of particular groups of users. There exists not one, single, coherent neogeographic space, but a vast multiplicity of user-generated places specific to particular communities of interest. Such places, generated through countless bottom-up interactions of users who are widely dispersed among many physical locations, are virtual, constantly changing, and often have only tenuous linkages to material landscapes. In addition, by allowing the local to become global, neogeographies cross conventional geographic scales with ease. In this sense, neogeography is compatible with the Deleuze–Guattarian 'flat ontology,' or spatialities unmarred by the obfuscating effects of scale (Marston *et al*. 2005). This notion is demonstrated by the multiple, multifaceted examples in the book *Else/Where: Mapping New Cartographies of Networks and Territories* (Abrams and Hall 2006), which embody the relational view of space and palimpsest view of place.

Neogeography and theories of truth: epistemological anarchism in the making?

If neogeography has offered alternative, pluralist ontologies of space and place, it also has important epistemological consequences. In particular, as diverse groups of people with varying agendas harness the web to upload their own data and interpretations, conventional views of what constitutes valid knowledge and truth come into question. For example, groups utilizing neogeographic technologies are not likely to generate random samples of data, a criterion that lies at the heart of commonly accepted (i.e., positivist) definitions of 'the scientific method.' Rather, the digital inputs that define group-specific, user-generated virtual worlds are selected with the commonality of intent that unites them into communities of interest. User-generated maps of outstanding nudist sites, endangered bird observations, handicapped-accessible restrooms, optimal bike paths or camping locations, accident-prone roads, green buildings, good shoe stores, vegetarian restaurants, or where people were located on 11 September 2001 may not pass the traditional standards of journal peer review and may not correspond with the 'facts' at the 95% statistical confidence level, but their efforts yield results that are important and meaningful to their respective groups of contributors, creating a 'people's geography' quite distanced from the rarified world of academia. What, then, is 'true' when people, particularly nonexperts, generate their own data and stories to make sense of the world in ways that may be markedly at odds with the rigid criteria demanded by GIS experts?

Departing from the correspondence theory of truth, neogeography allows for epistemological communities to construct 'truths' relative to specific groups of users. What is 'true,' for example, of place-based stories uploaded by a group of students through their iPhones may be quite different from the truth of environmental activists seeking to define the habitat of an endangered minnow. Such a perspective clearly points to the social and spatial construction of knowledge and the criteria by which we judge it to be 'true'. Truth, in this context, is removed from the exclusive domain of the intellectual, freed from its moorings in the correspondence theory, and instead is generated by those to whom it is useful in pragmatic, everyday terms.

There are many alternatives to the correspondence theory of truth, including constructivist, coherence, pluralist, performative, and pragmatist perspectives, all of which, to varying degrees, depart from conventional (i.e., orthodox positivist or 'scientific') perspectives by pointing to the

social origins and applications of knowledge. Whereas far from constituting some homogenous whole, these perspectives share a notion of 'truth' as something rather more complex than an unbiased representation of objective reality independent of the observer and applicable in all times and places. As Gibson-Graham (2000, p. 105) notes, 'If knowledge is not assigned the task of providing an accurate reflection of reality . . . then research doesn't simply reveal "what's out there" in the world.' Instead truth is considered as a partial, authored, and embodied series of intertwined statements created by diverse groups of people with widely different incentives, for numerous contrasting purposes, in specific and widely varying historical and geographic contexts (Livingstone 2003, Finnegan 2008). In this reading, knowledge is always embodied and spatially situated. As Hardy (2010) demonstrated, for example, even the number of contributors for Wikipedia entries follows a distance-decay function – people tend to contribute more information about places closer to where they live. However, in neogeography, the sharp divisions between the knower and the known, representations and the world they portray, epistemology and ontology are deliberately and creatively blurred, and truths (for there is always more than one) are repositioned as a partial, contingent series of statements that reflect lived reality and are useful in it. This line of thought owes much to John Dewey's and particularly William James's (1907, 1909) pragmatist epistemology, in which 'truth' is determined and confirmed by its utility and effectiveness in application, that is, from its consequences; thus, 'the "true" is only the expedient in our way of thinking, just as the "right" is only the expedient in our way of behaving' (James 1909, p. 2). Rather than the panopticonic gaze of professional GIS practitioners, neogeography has facilitated a democratization of truth production, transferring its creation and application from experts to the masses. To explicate this point, the works of three important philosophers are noted here: Paul Feyerabend, Jürgen Habermas, and Richard Rorty.

Feyerabend (1975) was instrumental in dethroning the correspondence theory of truth, advocating what amounted essentially to an anarchist theory of knowledge and ontological pluralism. In this line of thought, which preceded and facilitated the emerging discourses of postmodernism, there is no single observable reality or one methodological style. Rather, there are multiple realities, many different and equally viable ways of constructing knowledge, all of which require an enormous tolerance for inconsistency, incompleteness, and uncertainty. The parallels between Feyerabend's views and the perspectives of applied neogeographers are difficult to deny: both hinge on the acceptance of diverse, incommensurable ways of knowing, an acknowledgement of the multiple ways in which knowledge and social practices are intertwined, and a democratic celebration of very different bodies of knowledge outside the purview of self-appointed experts.

Another important contribution, and one of considerable relevance to neogeography, is Habermas's consensus theory of truth. As Habermas (1989) argues, communications are central to the social process of consensus and truth construction, through which individuals and communities of interest partake in the public, discursive interpretation of reality (cf. Calhoun 1992). Habermas's 'ideal speech situation' consisting of unfettered discourse is central to the 'public sphere' in which social life is constructed and reproduced and through which truth is constructed in the absence of barriers to communication. Truth, interpreted in this way, is inseparable from lived experience, intent, and social practice. In this reading, all participants in a debate theoretically have equal rights and abilities to make their views known and to challenge any other view. When all power relations have been removed from the freedom to engage in discourse, the only criteria for resolving contesting claims is their truth-value; importantly, 'the participants in an ideal speech situation [must] be motivated solely by the desire to reach a consensus about the truth of statements and the validity of norms' (Bernstein 1995, p. 50).

If one adopts a Habermasian approach in which democracy is approximated by an ideal speech situation' of unfettered discourse in the public sphere, neogeography provides a reasonably good approximation. Thus, neogeography practitioners can actively provide input into geographic representations, challenge dominant views, and mold the outcomes toward their own ends (Coleman *et al*. 2009, Graham 2009, Liu and Palen 2010). Accordingly, neogeography makes it possible for grassroots and nongovernmental organizations to shape the technology to their own needs (Sieber 2000). The explosion of cellular phones and similar portable devices has greatly facilitated the ability of many groups to upload and download neogeographic data (Ling and Campbell 2009). As Schuurman (2009, p. 571) observed, 'We face scenarios, previously unimaginable, of human beings as sensors gathering constant information with their cell phones and reporting back information to central, web based data collating sites.' Dunn (2007) cautions that participatory GIS, like neogeography, nonetheless faces real constraints to users, including the technical skills needed to upload data. Moreover, simply providing input is not the same as shaping policy outcomes that result from utilizing geodata. Of course, access to neogeography technologies is limited by social factors such as income, education, age, ethnicity, and gender, all of which are significant determinants of the 'digital divide' as well (Crang *et al*. 2006). Nonetheless, inasmuch as anyone with simple access to web tools to upload data and download results can participate in neogeographic communities, Habermas's ideal speech situation is largely realized for a vast potential pool of participants.

Similarly, Rorty (1979) offered a famous shift in metaphors for understanding knowledge. If the mirror served as the ideal 'reflection' of knowledge construction

from the Enlightenment onward (with its emphasis on vision, accuracy, and light), then the conversation serves as the perfect vehicle to summarize the forms of post-structuralist knowledge construction in the present era – a messy series of dialogues in which each voice is partial, incomplete, embodied, and contingent. In geography, a challenge to the hegemony of the visual has gradually occurred as the field has come to embrace social constructivist versions of truth rather than correspondence theories (Sui 2000). Neogeography is thus much more of a 'conversation' than it is a 'mirror' of the world in which the truths constructed are relative and useful for specific communities, there is no value-free arbiter to decide what is accurate/true, and accuracy is decided by consensus and pragmatic value. Moving away from the correspondence theory of truth, the conversational view puts more emphasis on practice, performance, and the speech acts that lead to points of agreement between contending worldviews (Strawson 1997). Neogeography has edged us closer to the performative theory of truth with an emphasis on actual spatial practices, that is, 'performing' these geographical or cartographic tasks on a sustained, repeated basis. Instead of truth, 'performativity' or 'truthiness' has been pushed to the front, that is, the quality of an idea 'being done right' or 'feeling true' without voluminous empirical evidence (Manjoo 2008). This line of thought has distinct, albeit as yet largely unselfconscious, parallels to nonrepresentational theory, with its emphasis on affect and prelinguistic, precognitive forms of knowledge (Thrift 2008).

Through selective exposure and peripheral processing, neogeographic media have enabled large numbers of people to pick and choose those sources of information that mesh conveniently with their ideological presuppositions. There are obviously both advantages and disadvantages to this approach. Most notably, it is not always the case that the 'wisdom of the crowd' is superior to that of a few experienced individuals. By utilizing data that only confirm their beliefs, users may never be confronted with disturbing or contradictory sources of information, which may reinforce existing stereotypes. In this sense, neogeography runs the risk of mirroring the highly specialized media markets in which viewers tend to utilize sources of news that conform to their political prejudices (Schuman 2004). However, by putting massive amounts of data and powerful visualization tools firmly in the hands of users and consumers rather than an elite group of expert producers, neogeography forces a broader recognition of the degree to which truth values reflect broader social concerns such as trust, reputation, and the credibility of volunteered information (Bishr and Kuhn 2007, Bishr and Mantelas 2008, Flanagin and Metzger 2008). Accuracy, in this context, is largely a matter of ideology and preference, contingent upon context and purpose, and tailored to the specific interests of communities of interest. In facilitating the emergence of numerous 'neighborhoods of truth,' neogeography encourages us to abandon the ridiculous holy grail of universal generalizations and come to terms with the place-bound nature of geographic knowledge. In its stead, it opens up venues for viewing the world through the eyes of particular groups bound together by lifestyle, political values, recreational habits, and other dimensions of social life.

Conclusions

The year 2010 marks the 20th anniversary for the coining of the term 'geographic information science' (Goodchild 2010). During the past 20 years, the GIScience community has made enormous progress in addressing the fundamental issues related to geographic data representation, analysis, and visualization (Goodchild 2009b). The explosive growth of neogeography in recent years poses new challenges and also offers new opportunities for both GIScientists and geographers. In this article, we aim to set the stage for a wider dialogue concerning the broader implications of neogeography, especially the ontological implications of neogeographical practices regarding the alternative conceptualizations of space and place and their consequences for theories of truth.

The recently released National Research Council (2010) report on the geographical sciences in the United States identified citizen mapping and mapping citizens as 1 of the 11 strategic directions on which geographers should focus in the near future. Citizen mapping and mapping citizens fit squarely within the scope of neogeography. The convergence of expert-led basic GIScience and geographic research with citizen-initiated neogeography is one area that likely will lead to exciting breakthroughs in GIScience and geography. As Downs (2010) recently observed, geography has been a popular subject throughout the history of the discipline, yet professional geographers have not done an adequate level of job of communicating their research to a broader audience. Web 2.0 technologies have brought the crowd of spatial data users to us through neogeography, and geographic knowledge is increasingly no longer the province of geographers and GIScientists, if it ever was. It is imperative for professional geographers and GIScientists to seize this unprecedented opportunity to make geography and GIScience more socially relevant by acknowledging the validity of user-generated communities of truth, the multiplicity of criteria that define useful knowledge, and the complex spatialities that arise from crowds who harness GIS for their own, rather than academic, purposes.

The popularity of neogeography reflects the increasingly fractalized nature of social and spatial life under contemporary capitalism, a world in which the structure and meaning of geography have been profoundly transformed under the simultaneous impacts of globalization and increasingly democratized information technologies.

Because it allows users to construct their own geographic representations, neogeography may be well suited to the intense time–space compression of this historical moment, or what Swyngedouw (1989) theorized as regions embedded in a volatile, postmodern 'hyperspace.' Location, in short, has become increasingly a matter of production and negotiation rather than being given *a priori*. If postmodern society exhibits a 'fibrous, thread-like, wiry, stringy, ropy, capillary character that is never captured by the notions of levels, layers, territories, spheres, categories, structures, or systems' (Paasi 2004, p. 541), then neogeography seems to be an ideal way for ordinary citizens to represent their geographies – the spatialities they construct and in which they live – to themselves, for themselves, and often by themselves.

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